

Discrete Latent Sequence Generation for 12-Lead ECG Reconstruction

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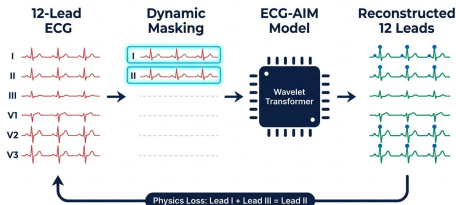
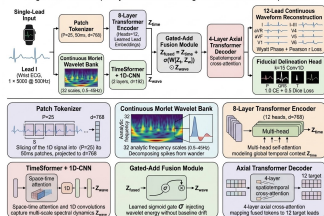
Clinical Background & Motivation

- ▶ **12-Lead Gold Standard:** Standard clinical cardiology relies on 10 electrodes capturing 3D cardiac electrical dipoles across frontal and horizontal planes, critical for diagnosing myocardial infarction, ischemia, and chamber hypertrophy.
- ▶ **Ambulatory Wearable Reality:** Consumer smartwatches (Lead I) and patch monitors (Lead II) provide continuous at-home monitoring but are blind to 10 of 12 spatial vectors.
- ▶ **The Inverse Problem:** Reconstructing 12 leads from limited leads is mathematically ill-posed; infinitely many 3D dipoles project to identical 1D traces.
- ▶ **Failure of MSE Regression:** Point-wise L_2 loss causes regression-to-the-mean, amplitude damping, and severe distortion of local clinical fiducials (QRS, ST, QT).

- ▶ **Novel Generative Framework:** Formulate **ECG-AIM**, a discrete latent sequence generation framework combining multi-resolution Morlet wavelets with spatiotemporal axial attention to reconstruct 12 leads from limited inputs.
- ▶ **Factorial 2^4 Loss Benchmarking:** Systematically dissect amplitude, correlation, distribution, and slope constraints across 3 paradigms (U-Net, MultiScale-VAE, ECG-AIM).
- ▶ **Physical Prior Enforcement:** Integrate Kirchhoff limb conservation ($I + III = II$) to ground frontal plane geometry.
- ▶ **Clinical & Hardware Validation:** Audit sub-10ms fiducial accuracy, downstream structural heart disease preservation, and zero-shot hospital cart transfer.

Methods: Architecture, Multi-Objective Loss & Training Pipeline

ECG-AIM Architecture (conv15e_A0_wave_noSSL_gated_add)
Single-Lead Time-Frequency Wavelet Conditioning & Multi-Task Clinical Supervision



1. Dual-Branch Model Architecture

Patch Tokenizer: 50 ms patches mapped to $d = 768$ temporal tokens via an 8-layer Transformer encoder.

Morlet Wavelet Bank: 32 analytic scales (0.5–45 Hz) extracts multi-resolution time-frequency scalograms.

Gated-Add Fusion: $Z_{\text{fused}} = Z_{\text{time}} + \sigma(W[Z_t, Z_w]) \odot \text{Proj}(Z_w)$.

Axial Decoder: Spatiotemporal cross-attention predicting 12-lead waveforms and explicit P-QRS-T masks.

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + 0.5\mathcal{L}_{\text{corr}} + 0.1\mathcal{L}_{\text{MMD}} + 0.05\mathcal{L}_{\Delta} + \lambda\mathcal{L}_{\text{SSL}}$$

Results and Conclusion: Clinical Precision & Hardware Translation



Endpoint Dimension	Baseline U-Net	Stanford 3DRECON	ECG-AIM (Ours)	Clinical Impact
Precordial Chest Mean r	0.570	0.732	0.776	$\Delta = +0.206$ ($p < 10^{-300}$)
QRS Duration MAE	13.85 ms	10.26 ms	8.0 ms (Median)	Sub-10ms clinical gate
Conduction Delay (>120 ms)	90.7%	94.3%	94.3%	aOR = 1.84 ($p < 10^{-17}$)
Sokolow-Lyon LVH Voltage	1.13 mV	0.63 mV	0.72 mV	$r = 0.714$ chamber concordance
EchoNext Structural Disease	68.8%	46.5% (Collapse)	71.6%	Preserves echo pathology

Leads I, II, and V_2 lock the 3 orthogonal dipoles (X, Y, Z), achieving precordial Pearson $r = 0.980\text{--}1.000$ and preserving anterior ST elevation.

Prior state-of-the-art (Stanford 3DRECON) collapsed to **46.5%** on downstream EchoNext structural heart disease. ECG-AIM maintains **71.6%** by preventing amplitude regression.